Engine Model(s): PT6A

72-50-05

EXHAUST DUCT - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

Not Applicable

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name
PWC32179	Base, Assembly

4. Fixtures, Equipment and Supplier Tools

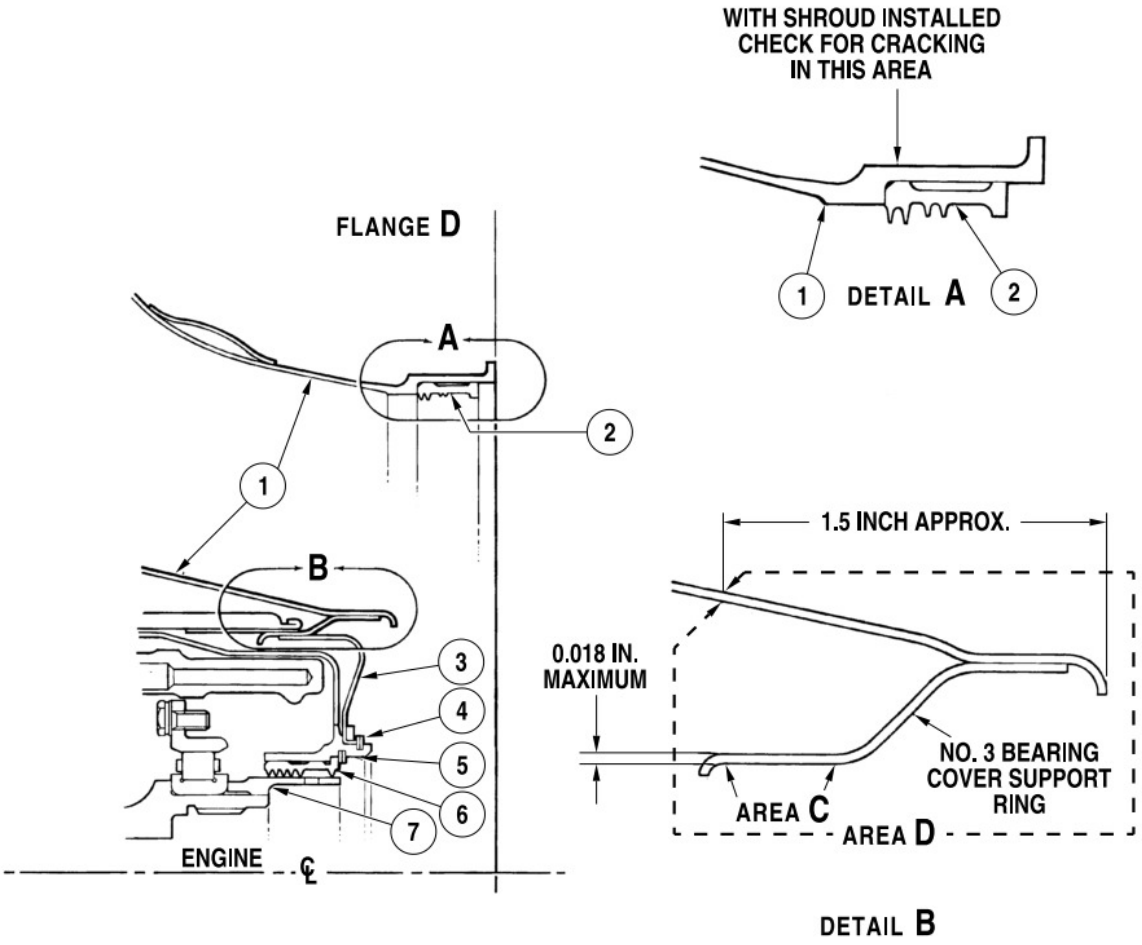
Not Applicable

5. Removal/Installation (Heavy Maintenance Only)

A. Removal of Exhaust Duct (Ref. Fig. 201)

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Figure 201 Removal/Installation of Exhaust Duct



Pre-SB1430

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1. Exhaust Duct
2. Power Turbine Shroud
3. No. 3 Bearing Cover
4. Bearing Cover Retaining Ring
5. Power Turbine Shaft Housing Assembly
6. Air Seal, Power Turbine Stator
7. Air Seal, Power Turbine Rotor

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(1) Remove the power section from the gas generator assembly (Ref. 72-50-03, Removal/Installation).

(2) With the base assembly (PWC32179) firmly secured to the workbench or other suitable fixture, position the power section vertically on the base, with the power turbine uppermost, and secure to the base using slave bolts.

(3) Remove the power turbine stator housing, power turbine vane ring, and interstage baffle (Ref. 72-50-03, Removal/Installation).

(4) Remove the power turbine disk and blade assembly (Ref. 72-50-04, Removal/Installation).

(5) Remove the retaining ring (4) and withdraw the No. 3 bearing cover (3) Pre-SB1430.

NOTE: The Post-SB1430 No. 3 bearing cover is an integral part of the exhaust duct assembly.

CAUTION: DO NOT SEPARATE THE REDUCTION GEARBOX FRONT AND REAR HOUSINGS AT FLANGE A.

(6) Remove the nuts, washers, bolts and lifting bracket at Flange A.

(7) Remove the exhaust duct assembly (1).

(8) Remove the thermal insulation blanket from the exhaust duct.

NOTE: Whenever the exhaust duct assembly is removed from the power section, inspect the reduction gearbox rear housing (Ref. 72-10-00).

(9) If the exhaust duct is cracked beyond limits (Ref. Para. 6.), remove the power turbine shroud (2) as follows:

(a) Apply a sufficient amount of dry ice to chill shrink the shroud diameter.

(b) Withdraw the power turbine shroud (2) evenly from the exhaust duct (1).

(c) Retain the shroud for installation in the replacement exhaust duct.

B. Installation of Exhaust Duct (Ref. Fig. 201)

(1) When the replacement exhaust duct is to be installed, install the power turbine shroud (2):

(a) To facilitate installation, chill shrink the shroud diameter using dry ice.

(b) Install the power turbine shroud (2) in the exhaust duct (1) with double knife edge lands toward the front of the engine. If the original shroud is being installed, align the "X" mark with the offset hole on the exhaust duct Flange D.

(c) Make sure the power turbine shroud (2) is properly seated.

(2) Install the thermal insulation blanket into the front of the exhaust duct assembly.

(3) Position the exhaust duct over the reduction gearbox rear housing.

(4) Position the lifting bracket (Ref. 72-00-00, REMOVAL/INSTALLATION). Secure Flange A using bolts (heads toward rear of engine), washers and self-locking nuts. Tighten the nuts, in diametric sequence, 36 to 40 lb.in.

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(5) Install the No. 3 bearing cover (3) Pre-SB1430 over the end of the power turbine shaft housing. Secure with the retaining ring (4).

NOTE: No. 3 bearing cover (Post-SB1430) is an integral part of the exhaust duct assembly.

CAUTION: WHEN POSITIONING THE POWER TURBINE DISK AND BLADE ASSEMBLY, EXERCISE CARE TO AVOID DAMAGING THE SHROUD LANDS BY CONTACT WITH THE BLADE TIPS.

(6) Install the power turbine blade and disk assembly (Ref. 72-50-04).

(7) Install the power turbine stator housing, vane ring and interstage baffle (Ref. 72-50-03).

6. Inspection/Check

The extent of power section disassembly will determine the scope of inspection to the exhaust duct.

A. Procedure

(1) For periodic, hot section, or unscheduled inspections such as engine sudden-stoppage (Ref. 72-00-00, INSPECTION, Unscheduled Inspection) or suspected aircraft heavy landing, examine the exhaust duct critically for evidence of the following:

(a) Examine the outer surface condition for buckling, ripples or similar distortion.

(b) Inspect outer surface, particularly in vicinity of flanges A and C for cracking in metal skin, welds, or flange bolt holes.

(c) Inspect exhaust port flanges for cracking. Cracks not exceeding 0.500 inch in length and do not progress into the stitch weld or cracks in a tangential direction not exceeding 1.000 inch long are acceptable provided they are stop drilled with a 1/16 (0.0625) inch drill.

(d) Check for the integrity of internal structure through the exhaust ports.

NOTE: Refer to the Aircraft Maintenance Manual for removal/installation of the exhaust stubs.

(e) Using suitable lighting or a borescope, examine the internal structure as far as possible for cracks, looseness and distortion.

(2) For engines that have incorporated SB1610 since the last overhaul and that exhibit inferior welds.

(a) Inspect for cracks at each minor inspection and repeat the inspection (Ref. Step (c)) at intervals not to exceed 150 hours of flight time.

(b) If cracks are found, repeat the inspection (Ref. Step (c)) at intervals not to exceed 25 hours flight time. All cracks will be marked with a suitable metal marking pencil (PWC05-018), and the length, location, duct hours and Time Since Overhaul recorded.

(c) Using 5x magnification, visually inspect the forward area of the exhaust duct for cracks, from the propeller reduction gearbox mounting flange to 2 inches aft around the entire circumference of the duct.

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1 Exhaust ducts are considered serviceable provided;

- There are no more than 3 cracks.
- The total length of all cracks does not exceed 2 inches.
- No one crack exceeds 1 inch.
- When there are 2 or 3 cracks they must be separated by a minimum of 6x the length of the longest crack or 3 inches, whichever is greater.
- The growth rate of any crack does not exceed 0.015 inch/hour of operation.

(3) With the power turbine vane ring and stator housing removed (Ref. Fig. 201):

(a) Inspect the exhaust duct (1) for cracks at Flange D in the location shown in detail A. Close visual and dye penetrant inspection should be carried out. When cracks are indicated, remove the exhaust duct for overhaul (Ref. Para. 5.A.).

(4) With the power turbine disk removed:

(a) Inspect the installed No. 3 bearing cover (3) Pre-SB1430. If evidence of looseness or rotation is present, refer to 72-50-04 Heavy Maintenance Only.

(b) Inspect the exhaust duct (1) area C where the bearing cover (3) Pre-SB1430 outside diameter mates with the exhaust duct. Light fretting may be polished out to the minimum material thickness of 0.018 inch. Evidence of the bearing cover rotation requires close visual and fluorescent dye penetrant inspection. The presence of cracks requires the removal of the exhaust duct for overhaul (Ref. Para. 5.A.).

(c) Inspect the end of the ski jump Pre-SB1430 area D, Detail B, for cracks using fluorescent dye penetrant inspection. No cracks allowed.

NOTE: After the exhaust duct has been removed, inspect the reduction gearbox rear housing (Ref. 72-10-00).

7. Cleaning/Painting

A. Procedure

(1) Remove salt and smog deposits from the outer surface of the exhaust duct with clean water. Dry with clean, dry compressed air.

(2) Clean light surface corrosion build-up with a non-metallic brush.

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